UIST 26 DETROIT

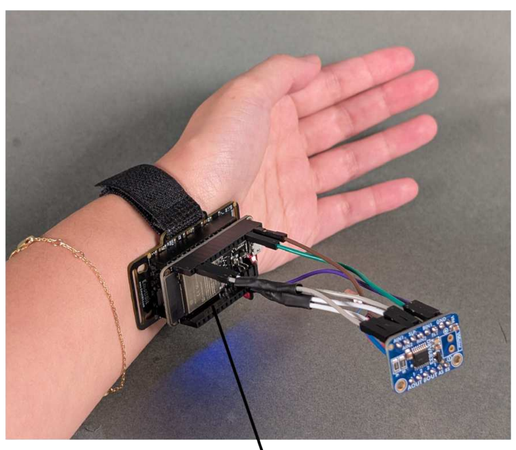
MIT CSAIL

CalmFold

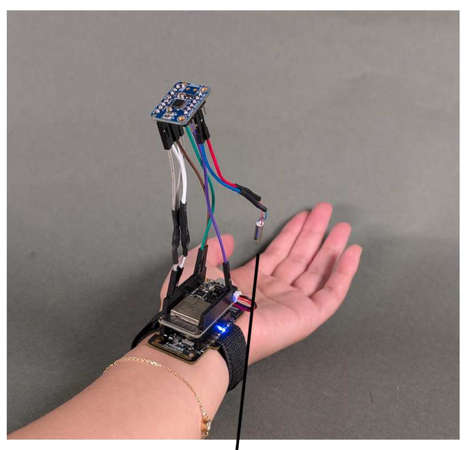
Feeling, Wearing, and Soothing Your Stress Through On-Body Physicalization

Kathy Nie and Emma Wu

Device Design

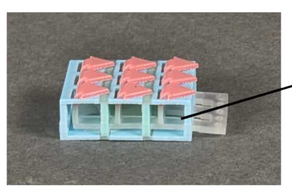


Emotibit senses the body's EDA to measure level of stress

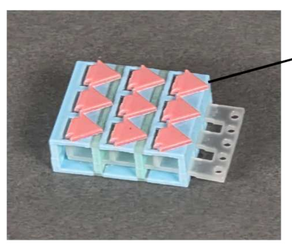


DC micromotor pulls on metamaterial

Metamaterial structure at rest

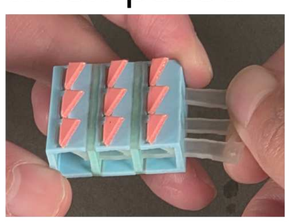


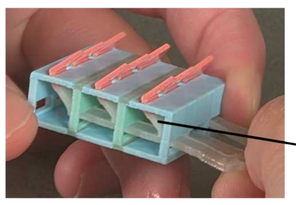
Inner structure



Spikes flush with top surface

Metamaterial structure when pulled





elastic pulling material

Metamaterial Structure has two states

System Design

EmotiBit

EmotiBit collects real-time EDA data from the user, which is analyzed to detect elevated stress and trigger the metamaterial.

Motor

A DC motor rotates a bobbin, pulling a tendon that actuates the metamaterial structure.

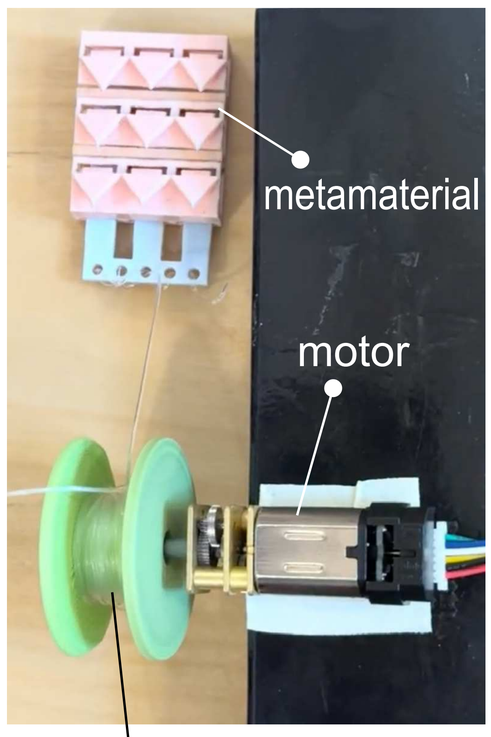
Metamaterial

When pulled, the metamaterial's spikes rise, providing tactile feedback that alerts the user that stress is elevated.

Stress Detection

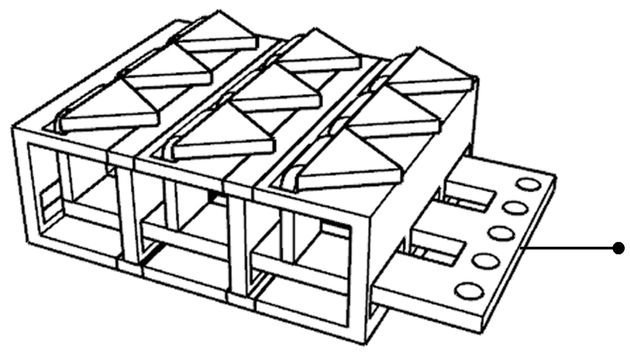


Actuation System



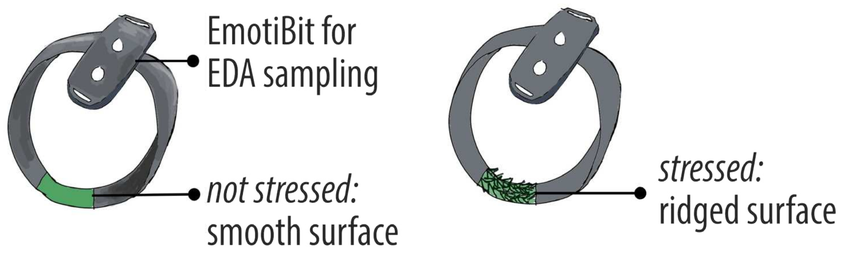
bobbin-shaped connector

Metamaterial Design



When handle is pulled, spikes will flip upwards

Calmfold physicalizes stress using metamaterial



metamaterial wrist band